

Computability: Exercise 12

Due: June 30th, 2026

Question 1

Claim: There exists a language L that can be recognized by a nondeterministic Turing machine in time n^4 but cannot be recognized by a deterministic Turing machine in place n^{10} .

Prove or disprove.

Question 2

We define a *log-space verifier* to be a deterministic Turing machine V with three tapes, operating on input pairs $\langle x, y \rangle$, where each component of the pair is given on a separate tape. The tapes are as follows:

- **Input tape:** A read-only tape containing the string x . The head may move in both directions.
- **Witness tape:** A read-only, one-way tape containing the string y . The head may move only to the right and cannot move left or stay in place.
- **Work tape:** A standard read-write tape used for computation, constrained to use $O(\log |x|)$ space (note that the space bound depends only on $|x|$, and not on $|y|$).

Prove that for a language $A \subseteq \Sigma^*$, $A \in \text{NL}$ if and only if there exists a log-space verifier V such that

$$A = \{x \in \Sigma^* : \text{there exists } y \in \Sigma^* \text{ s.t. } V \text{ accepts } \langle x, y \rangle\}.$$

Question 3

- a. We define the language **2-PATH** $:= \{\langle G, s, t \rangle : G \text{ is a directed graph, and there exist at least two different paths in } G \text{ from } s \text{ to } t\}$.

Prove that 2-PATH is NL-Complete.

Note: The paths may not be simple – the same vertex can appear more than once in every path.

- b. We define the language

$$2\text{SAT} := \{\langle \varphi \rangle : \varphi \text{ is a satisfiable 2CNF formula}\}.$$

Prove that 2SAT is NL-hard.

Hint: Use a reduction from $\overline{\text{PATH}}$ and Immerman theorem. The reduction assigns a variable for each vertex such that the assignment of a vertex indicates whether it is reachable from the source vertex.

Question 4

Let $A, B \in \text{NL}$, prove that:

- a. $A \cup B \in \text{NL}$.
- b. $A \cap B \in \text{NL}$.

Question 5

We define $E_{\text{DFA}} = \{\langle \mathcal{A} \rangle \mid \mathcal{A} \text{ is a DFA and } L(\mathcal{A}) = \emptyset\}$.

We define $A_{\text{DFA}} = \{\langle \mathcal{A}, w \rangle \mid \mathcal{A} \text{ is a DFA and } w \in L(\mathcal{A})\}$.

- a. Prove that if $3\text{SAT} \leq_L E_{\text{DFA}}$, then $\text{P} = \text{NP}$.
- b. Prove that if $E_{\text{DFA}} \leq_L A_{\text{DFA}}$ then $\text{L} = \text{NL}$.